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VEHICLE, VEHICLE CONTROL DEVICE, AND VEHICLE CONTROL METHOD

Abstract

A vehicle includes: a battery; an electric load; a power receiver having a power receiving coil that receives power from a power transmission coil installed on a road; a DC/DC converter that uses a switching element to convert the voltage value of DC power output from the power receiver; and a control device that controls the DC/DC converter. On the basis of predetermined conditions, the control device switches the states of the DC/DC converter when power is supplied from the power receiver to the battery and/or the electric load via the DC/DC converter between a non-operating state in which the switching element is maintained in the on state and an operating state in which the switching element is switched between the on and off states.

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Background/Summary

FIELD

[0001] The present invention relates to a vehicle, a vehicle control device, and a vehicle control method.

BACKGROUND

[0002] Conventionally, a technology for transmitting electric power in a contactless manner by using a transmission system such as a magnetic field resonance system is known (for example, JP 2013-191913 A). In such a technology, alternating-current power generated in a power-receiving coil by contactless power feeding is converted into direct-current power, and the voltage value of the direct-current power is converted by a DC-DC converter.

[0003] When such a technology is applied to contactless power feeding of a vehicle, electric power received by a power-receiving coil provided in the vehicle is supplied to a battery or the like through a DC-DC converter. At this time, an electric energy can be adjusted by the DC-DC converter, and stable power supply to the vehicle can be achieved.

SUMMARY

Technical Problem

[0004] However, when voltage conversion by a switching operation is performed every time electric power is supplied through the DC-DC converter, a power loss due to the switching operation occurs in the vehicle.

[0005] Then, in view of the aforementioned issue, an object of the present invention is to suppress a power loss in a vehicle while securing required electric power by power feeding when contactless power feeding to the vehicle is performed.

Solution to Problem

[0006] A summary of the present disclosure is as follows.

[0007] (1) A vehicle including: a battery; an electric load; a power reception device including a power-receiving coil receiving electric power from a power-transmitting coil provided on a road; a DC-DC converter converting, by using a switching element, a voltage value of direct-current power output from the power reception device; and a control device controlling the DC-DC converter, wherein, based on a predetermined condition, the control device switches a state of the DC-DC converter when electric power is supplied from the power reception device to at least one of the battery and the electric load through the DC-DC converter between a non-operating state in which the switching element is maintained in an on-state and an operating state in which the switching element is switched between an on-state and an off-state.

[0008] (2) The vehicle according to the aforementioned item (1), wherein the control device sets a

state of the DC-DC converter to the non-operating state when a predetermined received power reduction condition is satisfied.

[0009] (3) The vehicle according to the aforementioned item (2), wherein the received power reduction condition is that a length of a power feeding area in which the vehicle travels has a value equal to or greater than a predetermined value.

[0010] (4) The vehicle according to the aforementioned item (2), wherein the received power reduction condition is that an SOC of the battery has a value equal to or greater than a predetermined value.

[0011] (5) The vehicle according to the aforementioned item (2), wherein the received power reduction condition is a condition related to an electric energy consumed in the vehicle.

[0012] (6) The vehicle according to the aforementioned item (5), wherein the electric load includes a motor, and the received power reduction condition includes that power consumption of the motor has a value equal to or less than a predetermined value.

[0013] (7) The vehicle according to the aforementioned item (5) or (6), wherein the received power reduction condition includes that speed of the vehicle has a value equal to or less than a predetermined value.

[0014] (8) The vehicle according to any one of the aforementioned items (5) to (7), wherein the electric load includes an air conditioner, and the received power reduction condition includes that power consumption of the air conditioner has a value equal to or less than a predetermined value.

[0015] (9) The vehicle according to any one of the aforementioned items (5) to (8), wherein the electric load includes an air conditioner, and the received power reduction condition includes that outdoor temperature is within a predetermined range.

[0016] (10) The vehicle according to any one of the aforementioned items (1) to (9), wherein the control device sets a state of the DC-DC converter to the operating state when a predetermined power fluctuation inhibition condition is satisfied while electric power is supplied from the power reception device to the battery through the DC-DC converter.

[0017] (11) The vehicle according to the aforementioned item (10), wherein the power fluctuation inhibition condition includes that temperature of the battery is out of a predetermined range.

[0018] (12) The vehicle according to the aforementioned item (10) or (11), wherein the power fluctuation inhibition condition includes that each of electric power that can be charged to the battery and electric power that can be discharged from the battery has a value equal to or less than a predetermined value.

[0019] (13) A vehicle control device that controls a vehicle including: a battery; an electric load; a power reception device including a power-receiving coil receiving electric power from a power-transmitting coil provided on a road; and a DC-DC converter converting, by using a switching element, a voltage value of direct-current power output from the power reception device, wherein, based on a predetermined condition, a state of the DC-DC converter when electric power is supplied from the power reception device to at least one of the battery and the electric load through the DC-DC converter is switched between a non-operating state in which the switching element is maintained in an on-state and an operating state in which the switching element is switched between an on-state and an off-state.

[0020] (14) A vehicle control method for controlling a vehicle including: a battery; an electric load; a power reception device including a power-receiving coil receiving electric power from a power-transmitting coil provided on a road; and a DC-DC converter converting, by using a switching element, a voltage value of direct-current power output from the power reception device, the method including, based on a predetermined condition, switching a state of the DC-DC converter when electric power is supplied from the power reception device to at least one of the battery and the electric load through the DC-DC converter between a non-operating state in which the switching element is maintained in an on-state and an operating state in which the switching element is switched between an on-state and an off-state.

[0021] The present invention can suppress a power loss in a vehicle while securing required electric power by power feeding when contactless power feeding to the vehicle is performed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0022] FIG. 1 is a diagram schematically illustrating a configuration of a contactless power feeding system.

[0023] FIG. 2 is a diagram schematically illustrating a supply path of electric power in a vehicle.

[0024] FIG. 3 is a diagram illustrating an example of a configuration of a DC-DC converter.

[0025] FIG. 4 is a schematic configuration diagram of an ECU in a vehicle and equipment connected to the ECU.

[0026] FIG. 5 is a diagram illustrating an example of a power feeding area where a power-transmitting coil in a power feeding device is installed.

[0027] FIG. 6 is a flowchart illustrating a control routine of power reception processing.

DESCRIPTION OF EMBODIMENTS

[0028] An embodiment of the present invention will be described in detail below with reference to drawings. Similar components are given the same reference sign in the following description.

[0029] First, a configuration for supplying electric power to a vehicle in a contactless manner by using a power feeding device will be described. FIG. 1 is a diagram schematically illustrating a configuration of a contactless power feeding system **100**. The contactless power feeding system **100** includes a power feeding device **50** and a vehicle **1** and performs contactless power feeding between the power feeding device **50** and the vehicle **1**. In particular, the contactless power feeding system **100** according to the present embodiment performs contactless power feeding from the power feeding device **50** to the vehicle **1** by magnetic field resonant coupling (magnetic field resonance) when the vehicle **1** is traveling. In other words, the contactless power feeding system **100** transmits electric power from the power feeding device **50** to the vehicle **1** with a magnetic field as a medium. Contactless power feeding is also referred to as contactless power transmission, wireless power transmission, or wireless power feeding.

[0030] The power feeding device **50** is configured in such a way as to perform contactless power feeding to the vehicle **1**. Specifically, as illustrated in FIG. 1, the power feeding device **50** includes an electric power source **51**, a controller **52**, a communication device **53**, and a power transmission device **60**. The power feeding device **50** according to the present embodiment is provided on a road (lane) where the vehicle **1** travels and for example, is embedded underground (under the road surface). At least part of the power feeding device **50** (such as the electric power source **51**, the controller **52**, and the communication device **53**) may be placed above the road surface.

[0031] The electric power source **51** is an electric power source for the power transmission device **60** and supplies electric power to the power transmission device **60**. For example, the electric power source **51** is a commercial alternating-current electric power source supplying single-phase alternating-current power. For example, the electric power source **51** may be an alternating-current electric power source supplying three-phase alternating-current power.

[0032] The power transmission device **60** is configured to generate an alternating magnetic field for transmitting electric power to the vehicle **1**. The power transmission device **60** according to the present embodiment includes a power-transmission-side rectifier circuit **61**, an inverter **62**, and a power-transmission-side resonance circuit **63**. Suitable alternating-current power (high-frequency power) is supplied to the power-transmission-side resonance circuit **63** through the power-transmission-side rectifier circuit **61** and the inverter **62** in the power transmission device **60**.

[0033] The power-transmission-side rectifier circuit **61** is electrically connected to the electric

power source **51** and the inverter **62**. The power-transmission-side rectifier circuit **61** converts alternating-current power supplied from the electric power source **51** into direct-current power by rectification and supplies the direct-current power to the inverter **62**. For example, the power-transmission-side rectifier circuit **61** is an AC-DC converter.

[0034] The inverter **62** is electrically connected to the power-transmission-side rectifier circuit **61** and the power-transmission-side resonance circuit **63**. The inverter **62** converts direct-current power supplied from the power-transmission-side rectifier circuit **61** into alternating-current power (high-frequency power) with a frequency higher than that of alternating-current power of the electric power source **51** and supplies the high-frequency power to the power-transmission-side resonance circuit **63**.

[0035] The power-transmission-side resonance circuit **63** includes a resonator configured with a power-transmitting coil **64** and a power-transmission-side capacitor **65**. Various parameters of the power-transmitting coil **64** and the power-transmission-side capacitor **65** (such as the outside diameter and the inside diameter of the power-transmitting coil **64**, the number of turns of the power-transmitting coil **64**, and the capacitance of the power-transmission-side capacitor **65**) are determined in such a way that the resonance frequency of the power-transmission-side resonance circuit **63** is equal to a predetermined set value. For example, the predetermined set value is in a range of 10 kHz to 100 GHz and is preferably 85 kHz determined as a frequency band for contactless power feeding of a vehicle by the SAE TIR J2954 standard.

[0036] The power-transmission-side resonance circuit **63** is placed in the middle of a lane on which the vehicle **1** travels in such a way that the center of the power-transmitting coil **64** is positioned in the middle of the lane. When high-frequency power supplied from the inverter **62** is applied to the power-transmission-side resonance circuit **63**, the power-transmission-side resonance circuit **63** generates an alternating magnetic field for transmitting electric power to the vehicle **1**. The electric power source **51** may be a direct-current electric power source such as a fuel cell or a solar cell, and the power-transmission-side rectifier circuit **61** may be omitted in this case. Further, a filter circuit suppressing high-frequency noise generated from the inverter **62** may be provided between the inverter **62** and the power-transmission-side resonance circuit **63**.

[0037] For example, the controller **52** is a general-purpose computer and performs various types of control of the power feeding device **50**. For example, the controller **52** is electrically connected to the inverter **62** in the power transmission device **60** and controls the inverter **62** in such a way as to control power transmission by the power transmission device **60**.

[0038] The communication device **53** is equipment allowing communication between the power feeding device **50** and the outside of the power feeding device **50**. For example, the communication device **53** includes a short-distance wireless communication module (for example, a dedicated short-range communication (DSRC) antenna or a Bluetooth (registered trademark) module) for performing short-distance wireless communication and a wide-area wireless communication module for performing wide-area wireless communication. The communication device **53** is electrically connected to the controller **52**, and the controller **52** communicates with the vehicle **1** by using the communication device **53**.

[0039] On the other hand, the vehicle **1** includes a power reception device **2** and is configured in such a way as to be fed with power in a contactless manner by the power feeding device **50**. The power reception device **2** according to the present embodiment includes a power-reception side resonance circuit **21** and a power-reception side rectifier circuit **24**.

[0040] The power-reception side resonance circuit **21** is placed at the bottom of the vehicle **1** in such a way as to minimize the distance to the road surface. The power-reception side resonance circuit **21** according to the present embodiment is placed in the middle of the vehicle **1** in the width direction of the vehicle and is placed between the front wheels and the rear wheels in the longitudinal direction of the vehicle **1**.

[0041] The power-reception side resonance circuit **21** has a configuration similar to that of the

power-transmission-side resonance circuit **63** and includes a resonator configured with a power-receiving coil **22** and a power-reception side capacitor **23**. Various parameters of the power-receiving coil **22** and the power-reception side capacitor **23** (such as the outside diameter and the inside diameter of the power-receiving coil **22**, the number of turns of the power-receiving coil **22**, and the capacitance of the power-reception side capacitor **23**) are determined in such a way that the resonance frequency of the power-reception side resonance circuit **21** matches the resonance frequency of the power-transmission-side resonance circuit **63**. The resonance frequency of the power-reception side resonance circuit **21** does not necessarily match the resonance frequency of the power-transmission-side resonance circuit **63** as long as an amount of deviation between the resonance frequency of the power-reception side resonance circuit **21** and the resonance frequency of the power-transmission-side resonance circuit **63** is small, i.e., for example, the resonance frequency of the power-reception side resonance circuit **21** is within a $\pm 20\%$ range of the resonance frequency of the power-transmission-side resonance circuit **63**.

[0042] As illustrated in FIG. **1**, when an alternating magnetic field is radiated from the power-transmitting coil **64** while the power-receiving coil **22** in the power-reception side resonance circuit **21** is facing the power-transmitting coil **64** in the power-transmission-side resonance circuit **63**, the oscillation of the alternating magnetic field is transmitted to the power-reception side resonance circuit **21** resonating at the same resonance frequency as that of the power-transmission-side resonance circuit **63**. As a result, induced current flows through the power-receiving coil **22** in the power-reception side resonance circuit **21** by electromagnetic induction, and electric power is generated by the induced current. In other words, the power-receiving coil **22** receives electric power from the power-transmitting coil **64** provided on the road.

[0043] The power-reception side rectifier circuit **24** is electrically connected to the power-reception side resonance circuit **21**. The power-reception side rectifier circuit **24** converts alternating-current power supplied from the power-reception side resonance circuit **21** into direct-current power by rectification. For example, the power-reception side rectifier circuit **24** is an AC-DC converter. A filter circuit eliminating alternating-current power noise may be provided between the power-reception side resonance circuit **21** and the power-reception side rectifier circuit **24**.

[0044] FIG. **2** is a diagram schematically illustrating a supply path of electric power in the vehicle **1**. As illustrated in FIG. **2**, the vehicle **1** includes a DC-DC converter **3**, a relay **4**, a battery **5**, a sub-DC-DC converter **6**, an on-vehicle device **7**, a power control unit (PCU) **8**, and a motor **9** in addition to the power reception device **2**.

[0045] The DC-DC converter **3** is electrically connected to the power reception device **2** described above and specifically to the power-reception side rectifier circuit **24** in the power reception device **2**. FIG. **3** is a diagram illustrating an example of a configuration of the DC-DC converter **3**. As illustrated in FIG. **3**, the DC-DC converter **3** includes a switching element **31**, a diode **32**, a choke coil **33**, and a capacitor **34**.

[0046] By using the switching element **31**, the DC-DC converter **3** converts the voltage value of direct-current power output from the power reception device **2**. Specifically, the DC-DC converter **3** switches the switching element **31** between an on-state and an off-state and outputs voltage with a desired value by controlling a time ratio of the on-state to the off-state (a duty ratio). FIG. **3** illustrates a step-down DC-DC converter (a buck converter), and in this case, the DC-DC converter **3** steps down the voltage of input electric power. The DC-DC converter **3** may be a step-up DC-DC converter (a boost converter) or a step-up/down DC-DC converter (a buck-boost converter).

[0047] The relay **4** is placed between the DC-DC converter **3** and the battery **5** and controls an electric connection between the DC-DC converter **3** and the battery **5**. When the relay **4** is connected, i.e., when the relay **4** is in a closed state, electric power is supplied from the power reception device **2** to the battery **5** through the DC-DC converter **3**. On the other hand, when the relay **4** is broken, i.e., when the relay **4** is in an open state, power supply from the power reception device **2** to the battery **5** is broken.

[0048] The battery 5 accumulates electric power consumed in the vehicle 1. The battery 5 is a rechargeable secondary battery such as a lithium-ion battery or a nickel-hydrogen battery. When electric power is supplied from the power reception device 2 to the battery 5, the battery 5 is charged, and a state of charge (SOC) of the battery 5 is restored. Further, the battery 5 may be charged by an external electric power source other than the power feeding device 50 through a charge port provided in the vehicle 1.

[0049] The sub-DC-DC converter 6 is electrically connected to the DC-DC converter 3 and the battery 5 and is supplied with electric power from at least one of the power reception device 2 and the battery 5. The sub-DC-DC converter 6 steps down input voltage input to the sub-DC-DC converter 6 and supplies the stepped-down voltage to the on-vehicle device 7.

[0050] The on-vehicle device 7 is a part consuming electric power in the vehicle 1, examples of the device including an air conditioner, auxiliary machinery (such as an alternator, a water pump, and an oil pump), lighting equipment, and audio equipment.

[0051] The PCU 8 is electrically connected to the DC-DC converter 3 and the battery 5 and is supplied with electric power from at least one of the power reception device 2 and the battery 5. The PCU 8 includes an inverter and a step-up converter. The inverter converts direct-current power input to the PCU 8 into alternating-current power and supplies the alternating-current power to the motor 9. Further, the inverter converts alternating-current power generated by the motor 9 (regenerative electric power) into direct-current power and supplies the direct-current power to the battery 5. The step-up converter steps up direct-current voltage when regenerative electric power is supplied to the battery 5. The sub-DC-DC converter 6 and the PCU 8 may be integrally configured.

[0052] The motor 9 is an electric motor (such as an alternating-current synchronous motor) and is driven with electric power as a power source. The output of the motor 9 is transmitted to the wheels through a speed reducer and an axle. The vehicle 1 according to the present embodiment is a battery electric vehicle (BEV) not equipped with an internal combustion engine, and the motor 9 outputs power for traveling.

[0053] As illustrated in FIG. 2, the vehicle 1 according to the present embodiment includes the on-vehicle device 7 and the motor 9 as an electric load. Electric power received by the power reception device 2 is supplied to at least one of the battery 5 and the electric load through the DC-DC converter 3 depending on, for example, an open-close state of the relay 4.

[0054] FIG. 4 is a schematic configuration diagram of an electronic control unit (ECU) 10 in the vehicle 1 and equipment connected to the ECU 10. The vehicle 1 includes the ECU 10 as a control device of the vehicle 1. The ECU 10 executes various types of control of the vehicle 1.

[0055] As illustrated in FIG. 4, the ECU 10 includes a communication interface 11, a memory 12, and a processor 13. The communication interface 11, the memory 12, and the processor 13 are connected to each other through a signal line.

[0056] The communication interface 11 includes an interface circuit for connecting the ECU 10 to an on-vehicle network conforming to a standard such as a controller area network (CAN).

[0057] For example, the memory 12 includes a volatile semiconductor memory (such as a RAM) and a nonvolatile semiconductor memory (such as a ROM). The memory 12 stores a program executed by the processor 13, various types of data used when various types of processing are executed by the processor 13, or the like.

[0058] The processor 13 includes one or a plurality of central processing units (CPUs) and a peripheral circuit thereof and executes various types of processing. The processor 13 may further include an arithmetic circuit such as a logical operation unit or a numeric operation unit.

[0059] As illustrated in FIG. 4, the DC-DC converter 3, the relay 4, the sub-DC-DC converter 6, and the PCU 8 that are described above are electrically connected to the ECU 10. The ECU 10 controls power supply in the vehicle 1 by controlling each of the DC-DC converter 3, the relay 4, the sub-DC-DC converter 6, and the PCU 8.

[0060] The vehicle 1 further includes a global navigation satellite system (GNSS) receiver 14, a

map database **15**, a sensor **16**, a human machine interface (HMI) **17**, and a communication device **18**, which are electrically connected to the ECU **10**.

[0061] The GNSS receiver **14** detects the current position of the vehicle **1** (such as the latitude and longitude of the vehicle **1**), based on positioning information acquired from a plurality of (for example, three or more) positioning satellites. Specifically, the GNSS receiver **14** captures a plurality of positioning satellites and receives radio waves transmitted from the positioning satellites. Then, the GNSS receiver **14** calculates the distance to a positioning satellite, based on the difference between the time of transmission and the time of reception of radio waves and detects the current position of the vehicle **1**, based on the distances to the positioning satellites and the positions (orbit information) of the positioning satellites. The output of the GNSS receiver **14**, i.e., the current position of the vehicle **1** detected by the GNSS receiver **14** is transmitted to the ECU **10**.

[0062] The map database **15** stores map information. The map information includes positional information of a power feeding area to be described later, or the like. The ECU **10** acquires the map information from the map database **15**. The map database may be provided outside the vehicle **1** (for example, in a server), and the ECU **10** may acquire the map information from outside the vehicle **1**.

[0063] The sensor **16** detects a state of the vehicle **1** or an area around the vehicle **1**. The sensor **16** according to the present embodiment includes a vehicle speed sensor detecting the speed of the vehicle **1**, an outdoor temperature sensor detecting the outdoor temperature, a battery temperature sensor detecting the temperature of the battery **5**, the battery current sensor detecting input-output current of the battery **5**, or the like. The output of the sensor **16**, i.e., the state of the vehicle **1** or the area around the vehicle **1** detected by the sensor **16** is transmitted to the ECU **10**.

[0064] The HMI **17** performs input and output of information between the vehicle **1** and a crew of the vehicle **1** (such as a driver). For example, the HMI **17** includes a display displaying information, a speaker generating a sound, an operation button, an operation switch, or a touch screen for the crew to perform an input operation, and a microphone receiving a voice of the crew. The output of the ECU **10** is transmitted to the crew through the HMI **17**, and input from the crew is transmitted to the ECU **10** through the HMI **17**. The HMI **17** is an example of an input device, an output device, or an input-output device.

[0065] The communication device **18** is equipment allowing communication between the vehicle **1** and the outside of the vehicle **1**. For example, the communication device **18** includes a short-distance wireless communication module (for example, on-vehicle dedicated short range communication (DSRC) equipment or a Bluetooth (registered trademark) module) for performing short-distance wireless communication and a wide-area wireless communication module (for example, a data communication module (DCM)) for performing wide-area wireless communication. The ECU **10** communicates with the power feeding device **50** by using the communication device **18**.

[0066] FIG. **5** is a diagram illustrating an example of a power feeding area where the power-transmitting coil **64** in the power feeding device **50** is installed. Three power-transmitting coils **64** are spaced on the same lane of a road along the traveling direction of the vehicle **1** in the example in FIG. **5**. An area on a lane where a plurality of power-transmitting coils **64** are continuously installed corresponds to the power feeding area.

[0067] For example, when the vehicle **1** approaches the power feeding area, the ECU **10** transmits a power feeding request signal for requesting power feeding to the vehicle **1** to the power feeding device **50** by using the communication device **18**. When receiving the power feeding request signal from the vehicle **1**, the controller **52** in the power feeding device **50** generates an alternating magnetic field for power transmission by the power transmission device **60**. In other words, when receiving the power feeding request signal from the vehicle **1**, the controller **52** starts contactless power feeding from the power feeding device **50** to the vehicle **1**.

[0068] When electric power is transmitted from the power-transmitting coil **64** in the power transmission device **60** to the power-receiving coil **22** in the power reception device **2**, an electric energy is adjusted by a switching operation of the DC-DC converter **3** provided in the vehicle **1**. Thus, power can be stably supplied from the power feeding device **50** to the vehicle **1**, and by extension, a required amount of power feeding to the vehicle **1** can be secured. However, when voltage conversion by the switching operation is performed every time electric power is supplied through the DC-DC converter **3**, a power loss due to the switching operation occurs in the vehicle **1**.

[0069] Then, based on a predetermined condition, the ECU **10** according to the present embodiment switches the state of the DC-DC converter **3** when electric power is supplied from the power reception device **2** to at least one of the battery **5** and the electric load through the DC-DC converter **3** between a non-operating state and an operating state. In the non-operating state of the DC-DC converter **3**, the switching element **31** in the DC-DC converter **3** is maintained in an on-state, and conversion of a voltage value by the DC-DC converter **3** is not performed. On the other hand, in the operating state of the DC-DC converter **3**, the switching element **31** in the DC-DC converter **3** is switched between the on-state and an off-state, and conversion of a voltage value (step-down or step-up) by the DC-DC converter **3** is performed.

[0070] Since the switching operation of the DC-DC converter **3** is suspended in the non-operating state of the DC-DC converter **3**, a power loss due to the switching operation does not occur. On the other hand, electric power can be stably supplied from the power-transmitting coil **64** in the power feeding device **50** to the power-receiving coil **22** in the vehicle **1** by adjusting an electric energy by the switching operation of the DC-DC converter **3** in the operating state of the DC-DC converter **3**. Accordingly, switching the state of the DC-DC converter **3** between the non-operating state and the operating state, based on the predetermined condition, enables suppression of a power loss in the vehicle **1** while securing required electric power by power feeding.

[0071] For example, the predetermined condition includes a received power reduction condition. The received power reduction condition is a condition for allowing reduction in an electric energy received by the power-receiving coil **22** in the vehicle **1** and is satisfied when reduction in an electric energy is allowed. The ECU **10** sets the state of the DC-DC converter **3** to the non-operating state when the received power reduction condition is satisfied and sets the state of the DC-DC converter **3** to the operating state when the received power reduction condition is not satisfied. Setting the state of the DC-DC converter **3**, based on the received power reduction condition, can suppress shortage of electric power in the vehicle **1** due to reduction in received electric power in the non-operating state of the DC-DC converter **3**.

[0072] Further, the predetermined condition includes a power fluctuation inhibition condition. The power fluctuation inhibition condition is a condition for inhibiting fluctuation in an electric energy supplied from the DC-DC converter **3** to the battery **5** and is satisfied when fluctuation in an electric energy is inhibited. In this case, the ECU **10** sets the state of the DC-DC converter **3** to the operating state when the power fluctuation inhibition condition is satisfied while electric power is supplied from the power reception device **2** to the battery **5** through the DC-DC converter **3**. In other words, even when the received power reduction condition is satisfied, the ECU **10** sets the state of the DC-DC converter **3** to the operating state when the power fluctuation inhibition condition is satisfied. Setting the state of the DC-DC converter **3**, based on the power fluctuation inhibition condition, can suppress, for example, degradation of the battery **5** due to electric power fluctuation.

[0073] A flow of the control described above will be described below with reference to a flowchart in FIG. **6**. FIG. **6** is a flowchart illustrating a control routine of power reception processing. The control routine is repeatedly executed by the ECU **10** at predetermined execution intervals.

[0074] First, in step **S101**, the ECU **10** determines whether the power reception device **2** is receiving electric power. For example, the determination is made based on positional information

of the vehicle **1** and the power feeding area, power feeding information transmitted from the power feeding device **50** to the vehicle **1**, and/or the output of a current sensor or a voltage sensor provided in the power reception device **2**. When the power reception device **2** is determined to be not receiving electric power, the control routine ends. On the other hand, when the power reception device **2** is determined to be receiving electric power, the control routine advances to step **S102**. [0075] In step **S102**, the ECU **10** determines whether a predetermined received power reduction condition is satisfied. When the length of the power feeding area is long, time to feed electric power to the vehicle **1** can be secured; and therefore, reduction in an electric energy supplied to the vehicle **1** per unit time is allowed. Therefore, for example, the received power reduction condition is that the length of the power feeding area on which the vehicle **1** travels has a value equal to or greater than a predetermined value. In this case, the received power reduction condition is satisfied when the length of the power feeding area on which the vehicle **1** travels has a value equal to or greater than the predetermined value and is not satisfied when the length of the power feeding area on which the vehicle **1** travels has a value less than the predetermined value. For example, the predetermined value is set in a range of 10 m to 200 m. For example, positional information of the power feeding area including the length of the power feeding area is stored in the map information in the map database **15**.

[0076] Further, when the SOC of the battery **5** does not run short, necessity to increase an amount of power feeding to the vehicle **1** is not so great. Therefore, the received power reduction condition may be that the SOC of the battery **5** has a value equal to or greater than a predetermined value. In this case, the received power reduction condition is satisfied when the SOC of the battery **5** has a value equal to or greater than the predetermined value and is not satisfied when the SOC of the battery **5** has a value less than the predetermined value. For example, the predetermined value is set in a range of 50% to 80% and is preferably set to 65%. For example, the SOC of the battery **5** is calculated by integrating input-output current of the battery **5** detected by the battery current sensor or is calculated by using a state estimation method such as a Kalman filter.

[0077] Further, the received power reduction condition may be a condition related to an electric energy consumed in the vehicle **1**. In this case, for example, the received power reduction condition includes that the power consumption of the motor **9** has a value equal to or less than a predetermined value. In other words, the received power reduction condition is satisfied when the power consumption of the motor **9** has a value equal to or less than the predetermined value and is not satisfied when the power consumption of the motor **9** has a value greater than the predetermined value. For example, the predetermined value is set in a range of 2 kW to 4 kW and is preferably set to 3 kW. For example, the power consumption of the motor **9** is calculated based on required torque and/or electric power supplied to the motor **9**.

[0078] Basically, an electric energy consumed in the vehicle **1** increases as the speed of the vehicle **1** increases. Therefore, the received power reduction condition may include that the speed of the vehicle **1** has a value equal to or less than a predetermined value as a condition related to an electric energy consumed in the vehicle **1**. In this case, the received power reduction condition is satisfied when the speed of the vehicle **1** has a value equal to or less than the predetermined value and is not satisfied when the speed of the vehicle **1** has a value greater than the predetermined value. For example, the predetermined value is set in a range of 15 km/h to 40 km/h and is preferably set to 20 km/h. For example, the speed of the vehicle **1** is calculated based on the output of the vehicle speed sensor.

[0079] Further, the received power reduction condition may include that the power consumption of the air conditioner has a value equal to or less than a predetermined value as a condition related to an electric energy consumed in the vehicle **1**. In this case, the received power reduction condition is satisfied when the power consumption of the air conditioner has a value equal to or less than the predetermined value and is not satisfied when the power consumption of the air conditioner has a value greater than the predetermined value. For example, the predetermined value is set in a range

of 0.3 kW to 1 kW and is preferably set to 0.5 kW. For example, the power consumption of the air conditioner is calculated based on setting information of the air conditioner (such as a set temperature and an airflow volume) input to the HMI **17** by a crew of the vehicle **1** and/or electric power supplied to the air conditioner.

[0080] Further, when the outdoor temperature is a comfortable temperature for a crew of the vehicle **1**, the air conditioner is not likely to be operated. Therefore, the received power reduction condition may include that the outdoor temperature is within a predetermined range as a condition related to an electric energy consumed in the vehicle **1**. In this case, the received power reduction condition is satisfied when the outdoor temperature is within the predetermined range and is not satisfied when the outdoor temperature is out of the predetermined range. For example, the predetermined range is set to 5° C. to 20° C. or 10° C. to 20° C. For example, the outdoor temperature is calculated based on the output of the outdoor temperature sensor or is acquired based on weather information transmitted to the vehicle **1** from outside the vehicle **1**.

[0081] All or part of the conditions described above may be used as a condition related to an electric energy consumed in the vehicle **1**. When a condition related to an electric energy consumed in the vehicle **1** includes a plurality of conditions, the received power reduction condition is satisfied when all of the plurality of conditions are satisfied, and the received power reduction condition is not satisfied when at least one condition out of the plurality of conditions is not satisfied.

[0082] When the received power reduction condition is determined to be satisfied in step **S102**, the control routine advances to step **S103**. In step **S103**, the ECU **10** determines whether electric power is supplied from the power reception device **2** to the battery **5** through the DC-DC converter **3**. For example, the determination is made based on the state of the relay **4** and/or the output of the battery current sensor. When electric power is determined to be supplied to the battery **5** in step **S103**, the control routine advances to step **S104**.

[0083] In step **S104**, the ECU **10** determines whether the power fluctuation inhibition condition is satisfied. When the temperature of the battery **5** is off a proper temperature, degradation of the battery **5** may be accelerated due to fluctuation in electric power supplied to the battery **5**. Therefore, for example, the power fluctuation inhibition condition includes that the temperature of the battery **5** is out of a predetermined range. In this case, the power fluctuation inhibition condition is satisfied when the temperature of the battery **5** is out of the predetermined range and is not satisfied when the temperature of the battery **5** is within the predetermined range. For example, the predetermined range is set to 0° C. to 45° C. For example, the temperature of the battery **5** is calculated based on the output of the battery temperature sensor.

[0084] Further, the power fluctuation inhibition condition may include that each of electric power that can be charged to the battery **5** and electric power that can be discharged from the battery **5** has a value equal to or less than a predetermined value. In this case, the power fluctuation inhibition condition is satisfied when each of electric power that can be charged to the battery **5** and electric power that can be discharged from the battery **5** has a value equal to or less than the predetermined value and is not satisfied when at least either of electric power that can be charged to the battery **5** and electric power that can be discharged from the battery **5** has a value greater than the predetermined value. For example, the predetermined value is set in a range of 2 kW to 4 kW and is preferably set to 3 kW. Electric power that can be charged to the battery **5** is calculated as the absolute value of allowable charged power W_{in} and for example, is calculated based on the SOC of the battery **5** and/or the temperature of the battery **5**. Electric power that can be discharged from the battery **5** is calculated as allowable discharged power W_{out} and for example, is calculated based on the SOC of the battery **5** and/or the temperature of the battery **5**.

[0085] When the power fluctuation inhibition condition is determined to be not satisfied in step **S104**, the control routine advances to step **S105**. When electric power is determined to be not supplied to the battery **5** in step **S103**, the control routine skips step **S104** and advances to step

S105. In step **S105**, the ECU **10** sets the state of the DC-DC converter **3** to the non-operating state. In other words, the ECU **10** maintains the switching element **31** in the DC-DC converter **3** in the on-state. The control routine ends after step **S105**.

[0086] On the other hand, when the received power reduction condition is determined to be not satisfied in step **S102** or the power fluctuation inhibition condition is determined to be satisfied in step **S104**, the control routine advances to step **S106**. In step **S106**, the ECU **10** sets the state of the DC-DC converter **3** to the operating state. In other words, the ECU **10** switches the switching element **31** in the DC-DC converter **3** between the on-state and the off-state according to a set value of the duty ratio. The control routine ends after step **S106**.

[0087] While a preferred embodiment according to the present invention has been described above, the present invention is not limited to the embodiment, and various modifications and changes can be made within the scope of the claims. For example, the vehicle **1** may be a hybrid electric vehicle (HEV) or a plug-in hybrid electric vehicle (PHEV) including an internal combustion engine and a motor as power sources for traveling.

[0088] Further, steps **S103** and **S104** may be omitted in the control routine of the power reception processing in FIG. **6**.

REFERENCE SIGNS LIST

[0089] **1** Vehicle [0090] **2** Power reception device [0091] **22** Power-receiving coil [0092] **3** DC-DC converter [0093] **31** Switching element [0094] **5** Battery [0095] **7** On-vehicle device [0096] **9** Motor [0097] **10** Electronic control unit (ECU) [0098] **50** Power feeding device [0099] **60** Power transmission device [0100] **64** Power-transmitting coil

Claims

1. A vehicle comprising: a battery; an electric load; a power reception device including a power-receiving coil receiving electric power from a power-transmitting coil provided on a road; a DC-DC converter converting, by using a switching element, a voltage value of direct-current power output from the power reception device; and a control device controlling the DC-DC converter, wherein, based on a predetermined condition, the control device switches a state of the DC-DC converter when electric power is supplied from the power reception device to at least one of the battery and the electric load through the DC-DC converter between a non-operating state in which the switching element is maintained in an on-state and an operating state in which the switching element is switched between an on-state and an off-state.
2. The vehicle according to claim 1, wherein the control device sets a state of the DC-DC converter to the non-operating state when a predetermined received power reduction condition is satisfied.
3. The vehicle according to claim 2, wherein the received power reduction condition is that a length of a power feeding area in which the vehicle travels has a value equal to or greater than a predetermined value.
4. The vehicle according to claim 2, wherein the received power reduction condition is that an SOC of the battery has a value equal to or greater than a predetermined value.
5. The vehicle according to claim 2, wherein the received power reduction condition is a condition related to an electric energy consumed in the vehicle.
6. The vehicle according to claim 5, wherein the electric load includes a motor, and the received power reduction condition includes that power consumption of the motor has a value equal to or less than a predetermined value.
7. The vehicle according to claim 5, wherein the received power reduction condition includes that speed of the vehicle has a value equal to or less than a predetermined value.
8. The vehicle according to claim 5, wherein the electric load includes an air conditioner, and the received power reduction condition includes that power consumption of the air conditioner has a value equal to or less than a predetermined value.

9. The vehicle according to claim 5, wherein the electric load includes an air conditioner, and the received power reduction condition includes that outdoor temperature is within a predetermined range.

10. The vehicle according to claim 1, wherein the control device sets a state of the DC-DC converter to the operating state when a predetermined power fluctuation inhibition condition is satisfied while electric power is supplied from the power reception device to the battery through the DC-DC converter.

11. The vehicle according to claim 10, wherein the power fluctuation inhibition condition includes that temperature of the battery is out of a predetermined range.

12. The vehicle according to claim 10, wherein the power fluctuation inhibition condition includes that each of electric power that can be charged to the battery and electric power that can be discharged from the battery has a value equal to or less than a predetermined value.

13. A vehicle control device that controls a vehicle including: a battery; an electric load; a power reception device including a power-receiving coil receiving electric power from a power-transmitting coil provided on a road; and a DC-DC converter converting, by using a switching element, a voltage value of direct-current power output from the power reception device, wherein, the vehicle is configured, based on a predetermined condition, to switch a state of the DC-DC converter when electric power is supplied from the power reception device to at least one of the battery and the electric load through the DC-DC converter, between a non-operating state in which the switching element is maintained in an on-state and an operating state in which the switching element is switched between an on-state and an off-state.

14. A vehicle control method for controlling a vehicle including: a battery; an electric load; a power reception device including a power-receiving coil receiving electric power from a power-transmitting coil provided on a road; and a DC-DC converter converting, by using a switching element, a voltage value of direct-current power output from the power reception device, the method comprising, based on a predetermined condition, switching a state of the DC-DC converter when electric power is supplied from the power reception device to at least one of the battery and the electric load through the DC-DC converter between a non-operating state in which the switching element is maintained in an on-state and an operating state in which the switching element is switched between an on-state and an off-state.
